

Solutions :

- Due to very large no. of particles.
- $a \rightarrow T = \text{constant}$
 $b \rightarrow P = \text{constant}$
 $c \rightarrow V = \text{constant}$
- Vibrational degree of freedom should be included.

$$4. \quad r_{\text{diffusion}} \propto \frac{1}{\sqrt{\rho}} \propto M^{-\frac{1}{2}} \left\{ \rho = \frac{M}{V} \right\}$$

$$6. \quad E \propto T \left\{ E = \frac{1}{2} KT \right\}$$

$$9. \quad \text{Mean free path } \lambda \propto \frac{1}{\sqrt{2} \pi n d^2}$$

$$n = \frac{N}{V} \quad n \rightarrow \text{decreases}$$

$$\lambda \rightarrow \text{increases}$$

$$10. \quad \text{Avg Kinetic Energy/molecule} = \frac{3}{2} KT$$

It does not depend on the mass of the molecule.

$$11. \quad \text{KE/molecule} = \frac{f}{2} KT$$

KE does not depend on pressure.

$$12. \quad \text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{Mg}{\text{Area of piston}} = \text{constant}$$

$$\text{as } P = \text{constant} \quad V \propto T$$

If temperature increases then volume increases.

- Because molecules are travelling with different speeds

$$14. \quad \bar{P} = m\bar{v}$$

as average velocity of molecule is zero hence average momentum of molecule will be zero.

$$\bar{P} = m\bar{v}$$

$$15. \quad v_{\text{rms}} = \sqrt{\frac{3KT}{m}}$$

and at constant volume $P \propto T$

$$16. \quad \text{Average energy in one dimension} = 2 \times \frac{1}{2} KT$$

In three dimension energy = $3KT$

for one mole energy = $3KT \times N_A = 3RT$

$$C = \frac{\Delta Q}{\Delta T} = \frac{dU}{\Delta T} = 3R$$

- We treat water like solid For each atom average energy = $3KT$

water molecule has three atoms

$$\therefore \text{Total energy } U = 3 \times 3 KT \times N_A = 9RT$$

$$C = \frac{\Delta Q}{\Delta T} = \frac{dU}{\Delta T} = \frac{3RT}{T} = 9R$$

$$18. \quad PV = nRT \quad \left(n = \frac{m}{M} \right)$$

$$PV = \frac{mRT}{M}$$

$$P = \left(\frac{mR}{MV} \right) T$$

$$\text{Slope} \propto \frac{m}{V}$$

mass is doubled and volume is halved, slope becomes four times. Line will have slope more than line-A

- Molecules of air moving at high speed as per kinetic theory of gas collide and exchange their energy continuously.
- Molecules of gas have a finite though small size, so they are bound to undergo collision. As a result, they cannot move straight unhindered, their path keep getting deflected.
- As per kinetic interpretation of temperature, average kinetic energy of gas molecules is measure of temperature of gas sample.
- It is used to predict specific heat for all.
- Adiabatic compression increases its internal energy which in turn increase the KE of gas molecules. Piston (moving part) is responsible for it.

24. As in an ideal gas the molecular interaction is negligible, the concept of potential energy of molecules is not applicable here
So, T.E.(U) = KE + PE
U = KE for ideal gas.
25. The prediction of specific heat is based on law of equipartition of energy as temp $T \rightarrow 0$, degree of freedom get frozen and ineffective at low temperature.
26. According to kinetic theory, molecules of a gas are in a state of continuous random motion when the container is placed on fast moving train, ordered KE of molecules increases, that do not converted into disordered KE so temperature of gas remains same.
27. Mayer's relation is true for all type of ideal gase.